

LLM Selection for the AI-Powered Robotic Proxy

Recommendation - Team Beta (AI Stack)

1. Benchmark Method

Llama 3.1 (8B), Mistral (7B), Phi-3 Mini (3.8B), and Qwen2 (7B) were tested via Ollama with a single fixed business-pitch prompt on a CPU-only PC (no discrete GPU) as a worst-case baseline.

Model	Tokens/sec (CPU)	Params	Notes
Phi-3 Mini	3.91 (fastest)	3.8B	Fastest and lightest; concise, on-topic answers
Qwen2	2.39	7B	Strong multilingual support; coherent business tone
Llama 3.1	2.20	8B	Best instruction-following and reasoning consistency
Mistral	1.89 (slowest)	7B	Slowest in test; no quality edge over the others

2. Impact of the 16GB GPU

Moving inference to a 16GB VRAM GPU removes the CPU bottleneck seen above. All four models fit fully in VRAM with headroom to spare, and generation speed should improve roughly 10-20x (commonly 30-80+ tokens/sec for 7-8B Q4 models on a modern 16GB GPU), making real-time streamed conversation feasible for any of them. At that point, model choice should be driven by response quality and reliability rather than raw speed.

3. Recommendation

Primary (remote GPU server): Llama 3.1 8B-Instruct - best instruction-following and consistent professional tone for representing a business.

Secondary / alternate: Qwen2 7B-Instruct - strongest multilingual support among the four; useful if expo visitors are not all English speakers.

Local edge fallback (Jetson, no GPU): Phi-3 Mini - smallest footprint and fastest even on weak CPU hardware in this benchmark; best for degraded/offline mode.

Not recommended as primary: Mistral - slowest in the benchmark with no measurable quality advantage for this use case.

4. Why This Fits the Project

This matches the remote-first, local-fallback architecture already defined for the robot proxy: the primary model runs on the 16GB GPU server for quality and speed, while Phi-3 Mini remains the Jetson-safe fallback model for network-degraded or offline operation, consistent with the Week 1 plan.